Constructing Robust Investment Strategies:

Regularization Techniques and CVaR Optimization

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BSE Summer School 2025 - Advanced Portfolio Management
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July 21, 2025

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1 Introduction

This report outlines some long-term investment strategies for a private client based in the United States with available capital of $100 million. The portfolios are constructed using optimization algorithms under specific constraints and aim to achieve an optimal balance between expected return and risk by leveraging diversification across multiple asset classes through Exchange Traded Funds (ETFs).

Specifically, the report develops portfolios on two main optimizations: mean-variance optimization and VaR-Constrained optimization. Initially, historical return data (monthly returns) is used, calculates the efficient frontier, and identifies the optimal Mean-Variance and VaR-constrained portfolios. Then, noting the overweight of some assets in portfolios with historical data, new optimal portfolio allocations are computed through Cross Validation and Ridge Regression.

The analysis also analyzes the estimation risk and makes considerations about the different investment strategies that emerge from the optimizations.

1.1 Client Profile

The client has the following characteristics:

- Type of investor: Private Investor
- Geographical location: United States
- Initial Net Worth allocated for investments: USD 100 million
- Strategy: Long-term - balancing risk and return

1.2 Portfolio Characteristics & Constraints

Key constraints include:

- Minimum of 20% allocation in bonds or money market instruments;
- No short sales allowed;
- Use of ETFs with at least 10 years of historical data;
- Investment across various asset classes: Bonds, Equities, Commodities, Precious Metals, Real Estate, and Money Market;
- A Value at Risk (VaR) limit: 1-month VaR at 99% confidence level must not exceed 3%;
- ETFs should be representative of their asset class, not selected for outperformance;
- Uses a limited number of ETFs to ensure clarity, cost-efficiency, and ease of communication with the client.

2 ETFs Selection and Diversification Analysis

2.1 ETFs Selection

To construct the portfolio, it's important to choose ETFs to represent the major asset classes. The selection was guided by:

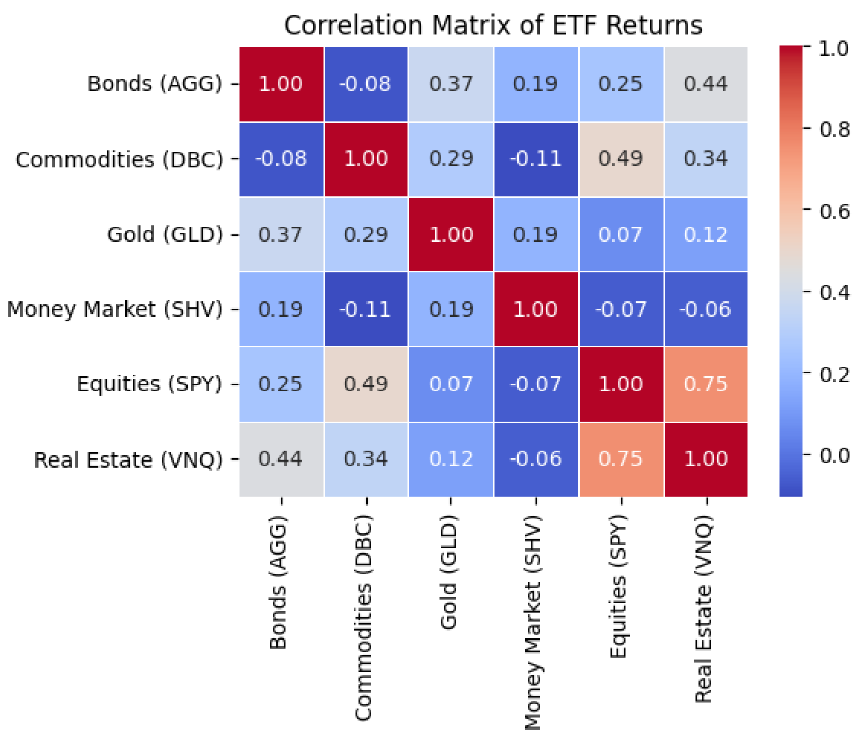
1. **Representativeness**: ETFs were chosen to represent their asset class, not to "beat the market."
2. **Sufficient History**: Only ETFs with at least 10 years of historical data were considered. This ensures the reliability of return and risk estimates, which are based on monthly data.
3. **Liquidity and Scale**: Preference was given to large, highly liquid ETFs with high trading volumes and large assets under management. These ETFs are widely used and provide cost effective access.

Ticker	Asset Class	ETF Name	Inception Year	Rationale
AGG	Bonds	iShares Core U.S. Aggregate Bond ETF	2003	Broad exposure to U.S. investment-grade bonds. Acts as a stabilizer with relatively low volatility.
SPY	Equities (U.S.)	SPDR S&P 500 ETF Trust	1993	Tracks the U.S. large-cap equity market. Most liquid ETF globally, widely recognized.
DBC	Commodities	Invesco DB Commodity Index ETF	2006	Provides diversified exposure to major commodities (energy, agriculture, metals). Useful in inflationary regimes.
GLD	Precious Metals	SPDR Gold Shares	2004	Physically backed gold ETF. Acts as a safe-haven asset and potential hedge against macro risks.
VNQ	Real Estate	Vanguard Real Estate ETF	2004	Provides access to the U.S. RE market. Adds diversification and potential income.
SHV	Money Market	iShares Short Treasury Bond ETF	2007	Ultra-short-term U.S. Treasuries. Very low risk, enhances liquidity and capital preservation.

2.2 Descriptive Statistics and Diversification Analysis

The ETFs shown above represent the benchmark for each asset class shown. To understand what contribution each asset class could make to the portfolio we look at the following statistics based on the last 18 years of data:

Asset Class	Expected Return (%)	Volatility (%)	Sharpe Ratio	Skewness	Kurtosis
Equities (SPY)	0.95	4.58	0.18	-0.58	3.82
Gold (GLD)	0.74	4.89	0.13	-0.00	3.24
Bonds (AGG)	0.24	1.37	0.10	0.32	5.83
Real Estate (VNQ)	0.74	6.54	0.10	-0.61	8.21
Money Market (SHV)	0.10	0.16	0.00	1.32	3.79
Commodities (DBC)	0.05	5.44	-0.01	-0.66	5.00



The selected ETFs have unique risk/return profiles and complementary behaviors that, together, increase the portfolio's diversification potential. The comparison of expected returns, volatility, and Sharpe ratios highlights the trade-offs between growth and stability inherent in each asset class.

Key findings:

- As expected, **SPY** (equities) has the highest expected long-term return and Sharpe ratio, but is characterized by high volatility, illustrating the underlying trade-off between risk and return in equity markets. This asset will certainly be a driver of the portfolio for long-term growth.

- In contrast, **AGG** (bonds) and **SHV** (money market) are low in volatility and act as portfolio stabilisers, protecting it during stock market downturns and contributing to capital preservation.
- **GLD** (gold) and **DBC** (commodities) are particularly useful for diversification, as they have a low or even negative correlation with equities (as we can see from the correlation matrix). This is especially true in periods of rising inflation or macroeconomic uncertainty, when conventional asset classes perform poorly. However, commodities have a very low expected return and relatively high volatility.
- **VNQ** (Real Estate) adds exposure to income-generating real estate assets with a risk/return profile between stocks and bonds, thus contributing to both return and diversification. However, note that this asset is highly correlated with stocks, but with a lower expected return and significantly higher volatility (the highest recorded in this data sample).

The distribution of each asset will then need to be analyzed for scenario simulations to be carried out later. In general, we see that except for SHV, which has high positive skewness, and VNQ, which has high kurtosis, the other securities have monthly returns distributed very close to a normal distribution.

In short, from the characteristics of each security and the correlation matrix, we see how these asset classes allow for a well-diversified portfolio, in which uncorrelated return flows minimize the overall volatility of the portfolio and improve risk-adjusted performance. By leveraging the unique economic sensitivities of equities, bonds, real assets, and commodities, investors can build a more durable and resilient allocation across a wide range of macroeconomic environments.

3 Portfolio Construction & Mean-Variance Frontier Analysis

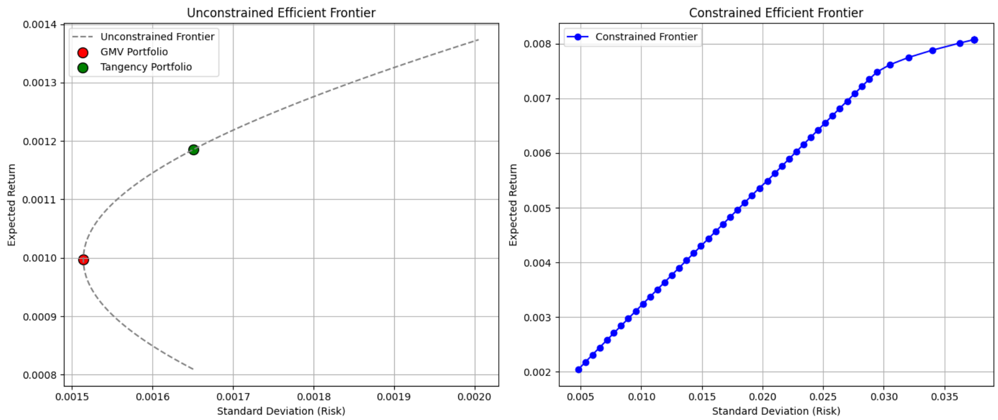
To find the efficient allocation of asset classes, we use the classic mean-variance model (Markowitz, 1952) and construct the efficient frontier with the following constraints:

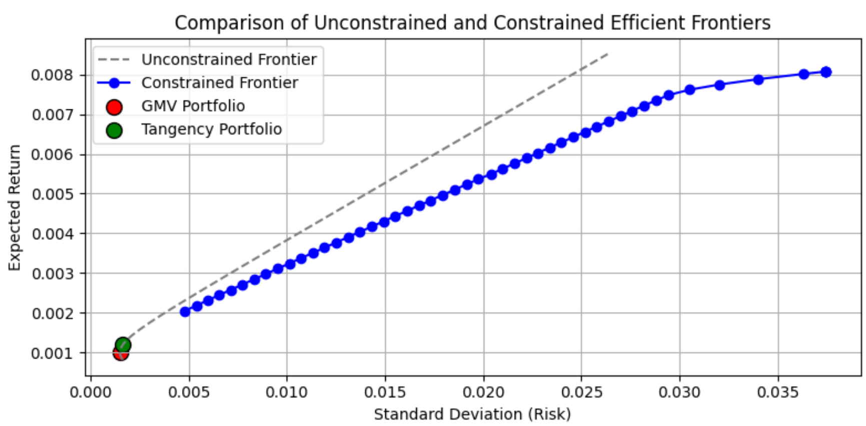
- No short selling (weights 0);
- Total investment (weights that add up to 1);
- Minimum allocation of 20% in bonds or money market (AGG + SHV 0.20).

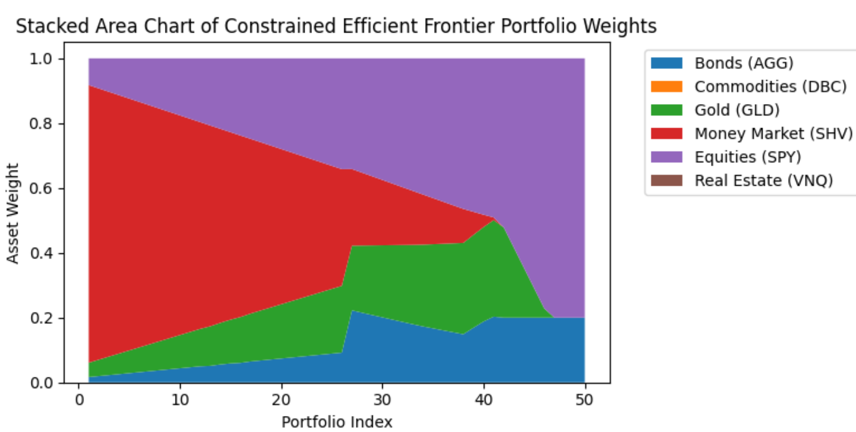
Three portfolios are calculated and analyzed:

1. **Global minimum variance portfolio (GMV):** the portfolio with the lowest possible risk (variance).
2. **Portfolio with maximum Sharpe ratio (tangency):** the portfolio with the optimal risk-adjusted return.
3. **Equal-weighted portfolio** for benchmarking.

The efficient frontier will also be constructed, which visually demonstrates how diversification improves return per unit of risk. The optimal portfolio according to Sharpe offers a significant improvement in expected return compared to GMV with a modest increase in volatility. In fact, the combination of the global minimum variance (GMV) portfolio and the tangency portfolio (maximum Sharpe ratio) offers the client a clear view of the risk-return spectrum.







The resulting **efficient frontier** succinctly demonstrates the fundamental trade-off between portfolio risk, measured by standard deviation, and expected return. Portfolios along the frontier are efficient combinations of assets that provide the maximum return available for a given level of risk.

In particular, we observe the **unconstrained and constrained frontiers**. As expected, the constrained frontier is significantly lower than the unconstrained frontier, reflecting the decrease in the set of achievable portfolios when short selling is not allowed and a minimum investment of 20% in bonds (AGG) and the money market (SHV) is imposed. Although these constraints ensure stability and compliance with regulatory and liquidity requirements, they reduce portfolio efficiency and limit the maximum return that can be achieved. In this case, the maximum expected return that can be achieved with the constraint is just over 0.86% (sd: 3.7%) per month, based on historical data. Without the constraint, higher returns could be achieved, but only at the expense of substantially higher volatility and downside risk.

The portfolio **weight chart** provides further insight into the optimiser's decisions. It is important to note that only equities (SPY), gold (GLD), bonds (AGG) and money market (SHV) are included in the efficient portfolios. The optimisation excludes real estate (VNQ) and commodities (DBC) along the frontier. This result indicates that these two asset classes, although they may offer diversification in certain contexts, were not sufficiently advantageous at the margin given the constraints. Real estate tends to have correlations with equities but offers lower risk-adjusted returns, making it less attractive in an optimised combination, as noted earlier. Commodities, on the other hand, have high volatility and only moderately negative correlations with other risky assets, making them less desirable in constrained portfolios that aim to optimise return and stability.

Along the efficient frontier, we observe the **portfolio with the highest Sharpe ratio**, which maximises risk-adjusted returns. This reflects the following dynamics:

- Expected Return of Maximum Sharpe Ratio Portfolio: 0.81%
- Standard Deviation of Maximum Sharpe Ratio Portfolio: 3.74%
- VaR (99%) of the Maximum Sharpe Ratio Portfolio: 1.43%
- CVaR (99%) of the Maximum Sharpe Ratio Portfolio: 2.05%

Maximum Sharpe Ratio Portfolio Weights:

Asset	**Weight**
Bonds (AGG)	2.86%
Commodities (DBC)	0.00%
Gold (GLD)	7.01%
Money Market (SHV)	77.72%
Equities (SPY)	12.40%
Real Estate (VNQ)	0.00%

In this case, the optimiser assigns a preponderant weight to the money market (SHV), a fairly substantial share to equities (SPY) and a minimal share to gold (GLD) and bonds (AGG). This portfolio appears to be very conservative, even though the expected return is not that low. In terms of risk, the standard deviation of the portfolio is significant considering the expected return, but VaR and CVaR, used to measure downside risk, are very low, proving the defensiveness of the portfolio.

Interpretation of MV optimisation for the client's choice. This portfolio would probably not be so suitable for long-term growth, but other portfolios along the constrained frontier could also be considered, ranging from highly conservative ones, dominated by the money market (SHV) and bonds (AGG), to more growth-oriented ones, with increasing weights in equities (SPY) and gold (GLD). For the client, a mid-to-high-end portfolio on the constrained frontier offers the optimal compromise between growth and stability. However, several improvements could be made to the optimisation model to better meet the client's needs.

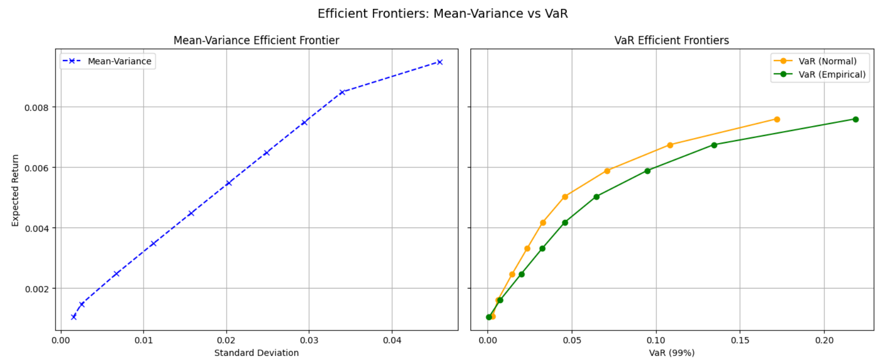
4 VaR-Constrained Optimization

To incorporate **downside risk management** into the portfolio construction process, we extend the analysis by identifying the portfolio with the highest expected return subject to a Value-at-Risk (VaR) constraint. Specifically, we impose a 1-month 3% VaR at a 99% confidence level, ensuring that the probability of experiencing losses greater than this threshold remains acceptably low. This approach addresses tail risk, which is particularly relevant for wealth management clients with strict risk tolerance or regulatory requirements.

To estimate risk more robustly, we generate additional return scenarios using both parametric methods, assuming normally distributed returns as in the mean-variance framework, and non-parametric bootstrapping. The latter captures potential deviations from normality, such as skewness and kurtosis, which may significantly affect extreme loss probabilities. Interestingly, apart from Money Market (SHV), Real Estate (VNQ) and Commodities (DBC), the selected assets exhibit skewness close to zero and kurtosis near three, suggesting an approximately normal distribution of returns. Nonetheless, using both methods provides a more comprehensive assessment of downside risk under different assumptions.

4.1 General analysis of optimisation with VaR constraints - Efficient frontier and weight graph

We then proceed with **VaR-Constrained optimisation** and show the efficient frontier and the weights assigned to the various assets by the optimiser.



The **efficient frontiers** highlight the trade-off between risk and return in the various optimisation strategies. Moving on to VaR-constrained optimisation, we see how the introduction of downside risk constraints modifies the frontier compared to the MV frontier, emphasising protection against the most severe losses.

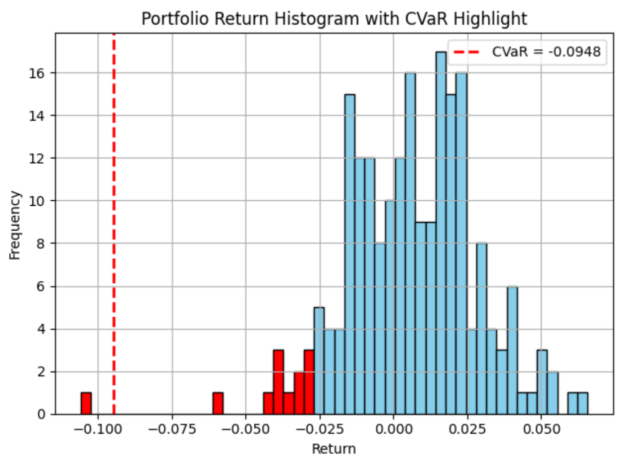
The contrast between frontiers estimated based on normality assumptions and those generated using empirical bootstrapping highlights estimation risk. The bootstrap frontier, although slightly less smooth, captures historical anomalies and is more representative of tail behaviour, providing a more conservative assessment of risk-return trade-offs.

The **weight chart**, on the other hand, offers further insight into how the optimiser reacts to these constraints. Within the VaR-constrained paradigm, portfolios show a strong preference for gold (GLD) and the money market (SHV), especially in conservative allocations, because these assets minimise exposure to market tail movements. As we increase the target return, weights migrate toward equities (SPY) to increase growth, while bonds continue to be underrepresented because they contribute minimally to risk-adjusted returns. The omission of real estate (VNQ) and commodities (DBC) indicates that these asset classes provide minimal incremental diversification in the context of the CVaR constraint.

This analysis confirms the choice of a top-segment portfolio on the constrained VaR frontier for the client, which balanced growth opportunities with strong downside protection.

4.2 Optimal portfolio analysis and interpretation of optimal allocation

The Value at Risk (VaR) and Conditional Value at Risk (CVaR) graph identifies potential losses for the portfolio in the event of extreme market scenarios. VaR establishes the maximum expected loss with a 99% confidence level over a one-month period, whereas CVaR quantifies the average loss exceeding this point, which offers a more complete indication of tail risk. Both measures underscore the necessity for including downside risk constraints in portfolio optimization. To better understand the concept of VaR and CVaR, we show the distribution of historical returns and highlight them on the **optimal portfolio**, the one with the highest Sharp Ratio.



The optimisation identifies an **optimal portfolio with VaR constraint** that balances return potential with downside risk protection. We note that the returns of this portfolio have a significant CVaR, due to a large left tail in the distribution.

In particular the Optimal VaR-Constrained Portfolio is characterized by: - Optimal Expected Return: 0.59% - Optimal Standard Deviation: 2.26% - Optimal VaR: 3.00% - Optimal CVaR: 9.48%

Optimal VaR-Constrained Portfolio Weights:

Asset	Weight
Bonds (AGG)	0.00%
Commodities (DBC)	0.00%
Gold (GLD)	31.67%
Money Market (SHV)	33.50%
Equities (SPY)	33.43%
Real Estate (VNQ)	0.00%

This optimal portfolio includes a decrease in expected return and a significant increase in CVaR, but the portfolio's standard deviation is more than 1% lower than the MV optimisation. Compared to the MV optimisation, the allocation to the money market is much smaller, while gold and equities gain greater weight. The portfolio takes on an almost equally weighted form between the three ETFs mentioned.

Interpretation of portfolio allocation with VaR constraint. Optimisation with VaR constraint assigns significant weight to gold (GLD). This is due to the historical function of this asset as a hedge in periods of market stress and its low correlation with equities (SPY). In cases of extreme downturns, gold has tended to preserve capital or even increase in value, making it extremely desirable in a risk-reduction context. This large allocation is also due to the fact that the model is based solely on historical data and may overestimate the past behaviour of gold in times of crisis.

CVaR figures around 10% seem high because they represent the average losses in the worst 1% of monthly results. Although such levels are not unusual for portfolios with significant exposure to equities, they can be reduced by lowering target returns, adding more bonds and cash, or extending the optimisation horizon to more than one month.

In summary, the VaR-constrained optimised portfolio reveals some fundamental limitations of the approach. Although the model may impose a strict monthly VaR limit of 3% at 99%, the resulting portfolio has a very high conditional value at risk (CVaR) of 9.48% and a reduction in expected return to 0.59% per month, which could severely penalise the portfolio's long-term performance.

From a **long-term asset management perspective**, this portfolio is suboptimal in terms of both risk-adjusted performance and diversification. In fact, assets such as bonds are neglected and the high weighting given to gold could be dangerous.

As an alternative to the VaR constraint, the explicit minimisation of CVaR should provide a better framework for controlling downside risk, as CVaR explicitly addresses losses at the tail of the return distribution. However, this is not the case. In fact, the minimisation of CVaR, the results of which we do not disclose here, leads to a further sacrifice of expected returns and increases the concentration in assets that are not sustainable for long-term growth.

These results indicate the need to align the optimisation objective with the client's actual risk tolerance. In practice, an intermediate portfolio that balances growth opportunities with tail risk management will be a more suitable choice for long-term wealth accumulation and preservation.

5 Estimation Risk Discussion and Model Improvement

One of the key challenges in portfolio optimization is **estimation risk** due to the uncertainty and possible inaccuracy of inputs to the model, mainly expected returns, volatilities, and correlations. Mean-variance and CVaR-constrained models put a lot of emphasis on historical data for estimating the parameters. But these estimates are noisy by nature and are susceptible to sampling variation, particularly if one takes a relatively short time horizon or if the asset return distributions have skewness and excess kurtosis.

In the current study, the VaR-optimizer's propensity for putting extreme weights, e.g., the overweighting of Gold (GLD), can be partly ascribed to estimation risk. Minor perturbations in estimated expected returns or covariances may cause the portfolio composition to change radically, as the optimizer attempts to take advantage of small perceived performance edges in the historical record. Likewise, the disparity between the frontiers arrived at under parametric (normality) assumptions and those produced by bootstrapping is indicative of the impact of estimation error and the existence of historical anomalies in the data.

To reduce estimation risk, the following improvements could be made:

- Resampling techniques to adjust for uncertainty in parameter estimates and generate more efficient frontiers that are robust.
- The Black-Litterman model, which blends equilibrium market returns and investors' views, can also moderate the sensitivity of portfolio weights to estimation error. This method produces allocations that are not just theoretically optimal but also more stable and in accord with practical investment concerns.
- Introducing Weight Constraints and Regularization, to prevent over-concentration in one specific asset (as with Gold in CVaR-restricted portfolios), weight constraints can be set in explicit terms or L2 regularization (ridge penalty) can be applied. This encourages diversification by penalizing large weight values.

It is important to address estimation risk so that portfolio recommendations are strong across various market regimes and do not depend too much on historical trends that might not continue going forward.

5.1 Incorporating Weight Constraints - Ridge Regression

To reduce estimation risk and to improve long-term portfolio weight stability, we added Ridge regression (L2 penalty) to the optimization. Ridge modifies return estimation by penalizing large coefficients in the regression formula, smoothing effectively the noise in the historical data and preventing extreme portfolio weights. This approach has several advantages: it distributes capital more evenly across a range of assets, does not over-adapt to past market idiosyncrasies, and creates portfolios that are less sensitive to changing market conditions. Ridge differs from Black-Litterman in that it applies only data-based regularization, without relying on subjective market expectations, which are often difficult to quantify and verify.

To improve the optimization model then the penalty term was introduced by calculating it through Cross Validation for each individual ETF. With these penalty terms the adjusted expected returns, called Ridge expected returns, were then calculated:

Ridge vs Historical Returns in percentages:

Ticker	Historical Return (%)	Ridge Expected Return (%)	Difference (%)
AGG	0.2379	0.3166	0.0787
DBC	0.0476	-0.1276	-0.1751
GLD	0.7371	0.5085	-0.2286
SHV	0.1043	0.0513	-0.0530
SPY	0.9495	1.0025	0.0531
VNQ	0.7419	0.9421	0.2003

Using the penalty-adjusted returns we can then redo MV and VaR-constrained optimization obtaining seemingly much more balanced and realistically usable results:

- The **Mean-Variance Ridge optimization** had an optimal portfolio with these characteristics:
  - Expected Monthly Return: 0.7879%
  - Portfolio Volatility: 1.5404%
  - Value-at-Risk (99%): 3.6869%
  - Conditional VaR (99%): 4.7546%
  - Optimal Weights: 64.32% Equities (SPY), 20.00% Bonds (AGG), 15.68% Gold (GLD).

  This allocation is consistent with a growth strategy with moderate downside protection. The relatively high exposure to Equities and Gold is indicative of Ridge's capacity to smooth expected return noise without over-dampening the exposure to riskier assets. Interestingly, though there was no explicit VaR constraint imposed, the portfolio 1-month VaR at 99% confidence is just below the 3% level, indicating an implicit risk discipline.

- The **VaR-constrained Ridge Optimization** had an optimal portfolio with these characteristics:
  - Expected Monthly Return: 0.8654%
  - Portfolio Volatility: 1.7390%
  - Value-at-Risk (99%): 0.0952%
  - Conditional VaR (99%): 1.3409%
  - Optimal Weights: 80.00% Equities (SPY), 20.00% Bonds (AGG).

  Here, optimization maximizes expected return under the strict 3% VaR constraint at 99% confidence. Surprisingly, the model assigns a very high weight to Equities in spite of the downside constraint. This is due to the interplay between Ridge regularization and the empirical return distribution: Ridge reduces volatility in expected returns, and the VaR constraint enables more aggressive investment in high-return assets if historical tail events are infrequent. Nevertheless, this allocation is risky in highly volatile markets since VaR does not account for the severity of losses beyond the threshold (CVaR is still non-negligible).

5.2 Comparative Insights & Considerations

The two Ridge-based optimization methods represent different trade-offs: - The Mean-Variance Ridge portfolio prioritizes diversified exposure with significant weightings to defensive assets (Gold and Bonds) to deliver smoother performance over market cycles. Although its VaR is marginally higher than the target, it provides an excellent trade-off between risk and return. - The VaR-Constrained Ridge portfolio, on the contrary, is more concentrated in Equities. Although the rigid VaR limit guarantees protection against tail losses at the given confidence level, it can generate overconfidence during periods of low volatility, possibly leaving the client exposed to unexpected tail risks not covered by VaR alone.

Note: The Ridge-optimal portfolios exhibit high expected returns with very low ex-ante risk measures (volatility, VaR, and CVaR). This is the direct result of Ridge regularization, which shrinks extreme return and covariance estimates towards their means. Consequently, risky assets like equities might seem more stable within the Ridge estimation framework, and the optimizer can allocate more weight to them without incurring large risk metrics. However, these numbers are forward-looking and represent the smoothed expectations of the model, not realized historical performance. When applied to historical data (ex-post), Ridge portfolios have a tendency to exhibit lower returns and higher risk than implied by the ex-ante estimates, since regularization prefers stability and robustness to fitting extreme market moves.

Therefore, let us examine these new portfolios and compare them to the previous ones to understand which is the best choice for the client's long-term strategy.

6 Comparative Analysis of Portfolio Strategies and Recommendation for the Client

Let us look at a comparison of the 4 optimal portfolios obtained from the four different optimizations:

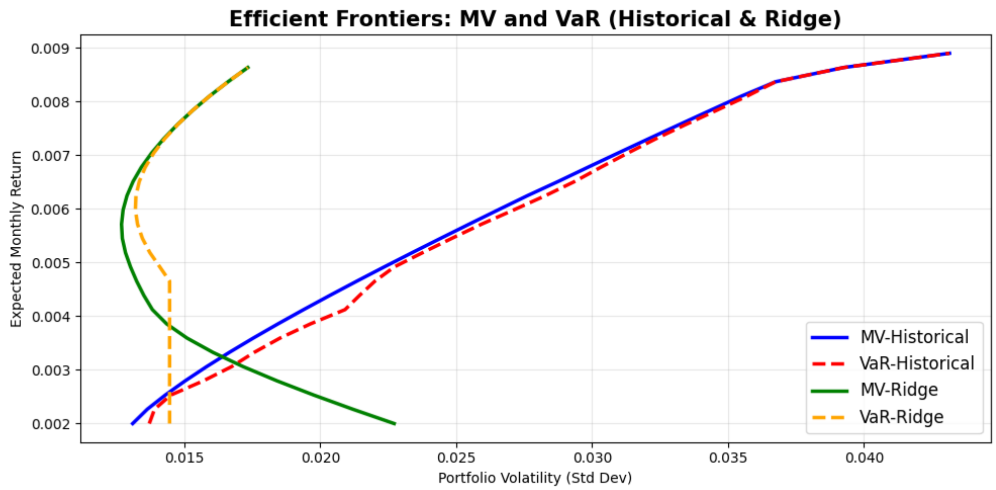
	MV-Historical	**VaR-Historical**	**MV-Ridge**	**VaR-Ridge**
Metrics (Ex-Ante):				
Expected Return	0.81%	0.59%	0.79%	0.87%
Volatility	3.74%	2.26%	1.54%	1.73%
VaR (99%)	1.43%	3.00%	3.69%	0.10%
CVaR (99%)	2.05%	9.48%	4.75%	1.34%
Weights:				
SPY (Equities)	12.40%	33.43%	64.32%	80.00%
AGG (Bonds)	2.86%	0.00%	20.00%	20.00%
GLD (Gold)	7.01%	31.67%	15.68%	0.00%
SHV (Money Market)	77.72%	33.50%	0.00%	0.00%
DBC / VNQ (Others)	0.00%	0.00%	0.00%	0.00%

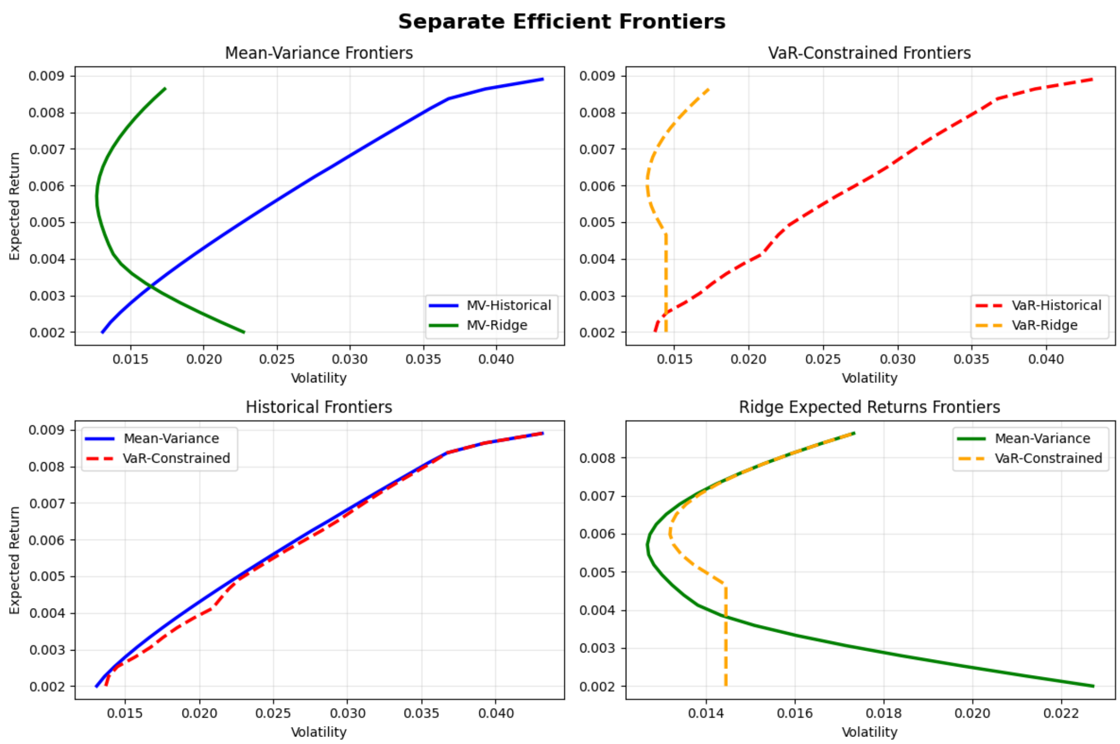
The summary table highlights the main trade-offs between return, volatility, and downside risk in the four optimization methods.

- The **MV-Historical portfolio** offers a very high expected return (0.81%), but makes extensive use of cash (SHV) and equities (SPY), resulting in moderate volatility (3.74%) and relatively low tail risk (CVaR 2.05%). However, this allocation may overlook the potential diversification benefits offered by other asset classes that would be necessary for long-term growth.
- The **VaR-Historical portfolio** achieves a significant reduction in volatility (2.26%), but at the cost of a significantly lower expected return (0.59%). Its CVaR rises sharply to 9.48%, indicating that, although the 3% Value-at-Risk constraint (99% confidence) is met, the portfolio remains vulnerable to extreme losses beyond this threshold. This therefore appears to be a suboptimal solution and is unlikely to be acceptable to the client, as it is too focused on highly volatile assets such as gold.
- The **MV-Ridge portfolio**, by regularizing return and covariance estimates, avoids extreme allocations and achieves a more balanced exposure (64% SPY, 20% AGG, 15% GLD). The ex-ante expected return (0.79%) is slightly lower than its historical counterpart, but the portfolio has more uniform risk characteristics (SD 1.54%, CVaR 4.75%) thanks to the Ridge penalty, which reduces the risk of overfitting and estimation risk.
- The **VaR-Ridge portfolio** is concentrated on only two assets: 20% bonds and 80% equities. Although it has the highest ex-ante expected return of all the portfolios (0.87%), it seems too concentrated on equities. From a downside risk perspective, it also appears to be an excellent portfolio, but we should remember what we said above: ex-post, the expected return of the portfolio is lower than that shown ex-ante, and the risk measures are also higher.

In general, the expected returns of portfolios optimized on historical data and those based on Ridge estimates are not directly comparable, and the same applies to risk measures (although these are more comparable). At first glance, portfolios constructed with Ridge-estimated returns appear to be significantly superior to historical ones in every respect. This discrepancy stems from Ridge's regularization effect, which returns a potentially overly optimistic view of portfolio efficiency. Despite this necessary disclaimer, Ridge portfolios are still more robust in practice and seem more suited to long-term investment strategies, as they prioritize stability and reduce the impact of estimation errors.

These considered are the optimal portfolios generated by the optimizations; however, depending on the client's choice and the utility the portfolio can provide him, other portfolios along the **efficient frontier** can also be chosen. We then look at the graphs of the efficient frontiers for the different optimizations.





The contrast between the four efficient frontiers provides an in-depth view of how each optimization technique balances risk and return.

Although both the historical frontiers and those corrected for Ridge Mean-Variance show the classic trade-off—an increase in risk leads to an increase in potential returns—the Ridge frontiers are slightly further to the left, indicating more diversified and stable allocations, a direct consequence of regularization that mitigates extreme weights.

Comparing the two optimization methodologies, the VaR-constrained frontiers are narrower and shifted further to the right, showing how the strict imposition of a Value-at-Risk constraint tightly circumscribes the set of available portfolios, as noted earlier. This constraint favors downside protection but reduces the potential for high returns. As might be expected, the Ridge-based VaR frontier is even more conservative than its historical equivalent, highlighting how the Ridge penalty rewards stability over adventurous positioning.

6.1 Final considerations and recommendations for the client

For the long-term asset management plan, the Mean-Variance Ridge solution seems more appropriate as it incorporates greater diversification and stability without having to rely on historical distribution assumptions on which VaR-constrained optimization must necessarily be based. However, the final choice of portfolio should take into account the client's risk tolerance and liquidity requirements, balancing capital appreciation with acceptable drawdown levels.

Both Ridge models generated more realistic and potentially feasible portfolio solutions in practice. By penalizing extreme allocations, Ridge regularization allocates capital more evenly across assets, avoiding high concentration in assets such as gold that do not typically account for a large portion of the portfolio allocation. This results in portfolios that are more in line with long-term asset management principles and less subject to distortions due to anomalies in historical data.

For the client, these observations underscore that while the mean-variance strategy offers higher potential returns, this comes at the cost of greater exposure to volatility and tail risks. Conversely, models with VaR constraints effectively limit downside risk but impose more severe constraints on achievable performance.

A conservative approach to portfolio construction would be to choose a portfolio along the Ridge Mean-Variance frontier that balances growth potential with stability, in line with the client's risk tolerance. However, since portfolio construction is personal, it is necessary to combine the client's wishes and risk appetite in deciding on the allocation that best suits their long-term financial goals.

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